

MANI VARDHAN SATHA

Embedded Firmware Engineer | IoT | Linux BSP

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PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 3 years of experience building production firmware on Qualcomm SM6112 and STM32 platforms, with shipped products in Industrial IoT and EV/solar charging. Proficient in Embedded C/C++, Linux BSP (Yocto/OpenWRT), FreeRTOS, device driver development, and edge AI deployment.

TECHNICAL SKILLS

Languages: Embedded C, C++, Python, Shell

Hardware Platforms: Qualcomm SM6112, STM32, ESP32, Raspberry Pi, ARM Cortex (M/A series)

Operating Systems: Embedded Linux (Yocto, OpenWRT), RTOS, FreeRTOS

Protocols: I2C, SPI, UART, CAN, BLE, MQTT, TCP/IP

Drivers & Low-level: Device Drivers, Interrupt Handling, DMA, Memory-mapped I/O, HAL, CMSIS, Bootloaders, Kernel Customization

Build & DevOps: CMake, Make, Cross-compilation, JTAG, GDB, Buildroot, Git, Logic Analyzer, CI/CD

AI & Computer Vision: OpenCV, YOLO, Deep Learning, Face Detection, Object Detection, Edge AI Deployment

Cloud & Other: AWS, MQTT, OTA Update

PROFESSIONAL EXPERIENCE

Embedded Software Engineer | Caribou Technologies, Hyderabad, India *Jun 2023 – Present*

- Led end-to-end firmware development for Industrial IoT systems in Embedded C, achieving measurable boot time optimization and real-time latency reduction across two product lines.
- Customized Linux kernel (BSP) on Qualcomm SM6112 (ARM) and Raspberry Pi; reduced wireless communication latency by **20%** and implemented watchdog recovery for high system uptime.
- Authored performance-critical Device Drivers (I2C/SPI/UART) with DMA and interrupt handling for 5+ environmental sensor nodes (CO₂, pressure, air quality), reducing data polling latency by **35%**.
- Developed and optimized proprietary Qualcomm Quick Charge (QC) profile algorithms using HAL/CMSIS abstractions; post-silicon validation boosted power efficiency by **15%**.
- Executed Linux kernel/network customization for OpenWRT systems (MIPS/ARM) including Python 2.7 → 3 migration and CMake/Makefile build system maintenance.
- Deployed a real-time Face Detection and Facial Recognition system using OpenCV and YOLO on edge devices via OTA update pipeline, achieving low-latency inference.

Embedded Software Engineer (Intern) | Caribou Technologies, Hyderabad, India *Dec 2022 – May 2023*

- Supported firmware bring-up and driver integration for sensor nodes; wrote unit tests using CppUTest to validate I2C/SPI driver behaviour.
- Assisted in OpenWRT build system setup (Buildroot/CMake), cross-compilation toolchain configuration, and JTAG/GDB-based debugging workflows.

TECHNICAL PROJECTS

Regenerative Hybrid Vehicle Control System · 2023

- Developed ARM Cortex MCU firmware for hybrid vehicle control logic; integrated Li-ion BMS and regenerative charging algorithms, reducing dual-powertrain state transition latency by ~40ms.

High-Efficiency Qi Wireless Solar Power Bank · 2023

- Designed PMIC firmware on a low-power MCU to manage Qi-standard wireless charging and solar acquisition profiles, achieving ~92% charging efficiency.

EDUCATION

B.E. in Electronics & Communication Engineering *May 2023*

Maturi Venkata Subba Rao (MVSRR) Engineering College, Hyderabad